DOES MACHINE LEARNING BEAT THE CREDIT SCORECARD? DISCRIMINATION, CALIBRATION AND PROFIT IN PROBABILITY-OF-DEFAULT MODELLING

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ABSTRACT

This report presents a self-guided term project on probability-of-default (PD) modelling, the central model of retail credit risk. Banks overwhelmingly deploy logistic-regression scorecards, while the machine-learning literature reports that gradient-boosted trees dominate them; the literature, however, scores models almost exclusively by discrimination (AUC), whereas a lender needs calibrated probabilities and profitable accept/reject decisions. This project tests whether the ML advantage survives on the criteria that matter. An industry-style weight-of-evidence scorecard and three challengers (XGBoost, LightGBM, a multilayer perceptron) were built on 621,022 Lending Club loans under a strict out-of-time protocol (train ≤2013, calibrate on the 2014 vintage, test on the 2015 vintage; origination-time features only) and evaluated on discrimination (AUC, KS), calibration before and after Platt and isotonic recalibration (Brier, ECE), and realized profit across accept/reject cutoffs. The boosted trees win, but narrowly and unevenly: XGBoost beats the scorecard by 0.85 AUC points (DeLong $p < 10^{-54}$), the edge persists after both models are recalibrated, and it buys 2.1% higher profit at the optimal cutoff; the MLP is significantly *worse* than the scorecard, and the scorecard is the best-calibrated model by ECE both before and after recalibration. A robustness run on the UCI Taiwan credit-card dataset reproduces the ranking. Well-tuned boosted trees earn their complexity; the forty-year-old scorecard covers most of the distance.

Index Terms: Credit scoring, probability of default, calibration, XGBoost, weight of evidence, profit-based evaluation

1. INTRODUCTION

When a bank decides whether to grant a loan, the decision rests on a single model output: the applicant's probability of default (PD). PD models sit at the heart of retail banking; they set interest rates and credit limits, drive the accept/reject decision, and, under the Basel accords and IFRS 9, determine regulatory capital and loan-loss provisions [1]. Because capital requirements scale with estimated PDs, these are arguably the highest-stakes statistical models in finance, and regulators require them to be accurate, stable and explainable.

The industry standard is remarkably old technology: a logistic regression fit on weight-of-evidence (WoE) transformed features, packaged as an additive point scorecard [3, 2]. Over the past decade, a large benchmarking literature has reported that machine-learning models, particularly gradient-boosted tree ensembles, outperform logistic regression on credit data [4, 5]. Yet almost all of that evidence is stated in terms of discrimination, the ability to rank bad borrowers above good ones, summarized by AUC. Discrimination is not what a lender ultimately consumes. Provisioning and capital need the PD itself to be right on average (calibration), and the lending decision needs the model to make money at an operating cutoff (profit). A model can gain AUC while producing badly miscalibrated probabilities, and a calibrated but weakly discriminating model can approach a stronger model's profit at the right cutoff.

This project puts the scorecard-versus-ML question to a controlled test on the criteria that matter. A single pipeline builds an industry-style WoE scorecard and three ML challengers on identical origination-time information from real consumer loans, evaluates them out of time (later vintages than training), applies identical post-hoc recalibration to every model, and translates every model's predictions into accept/reject profit under an explicit cash-flow assumption, so that any claimed ML advantage must survive calibration and decision analysis rather than an AUC comparison alone.

Section 2 formulates the problem and lists the objectives. Section 3 details the methodology: the data discipline, the scorecard, the challengers, calibration and the profit model. Section 4 presents the experimental setup and results. Section 5 discusses and interprets the findings, Section 6 concludes, and Section 7 links the project artifacts.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Let $(\mathbf{x}_i, y_i)$ describe loan i, where $\mathbf{x}_i$ contains only information available on the day the loan was issued and $y_i \in \{0, 1\}$ indicates whether the loan was eventually charged off. The task is **probability-of-default estimation**: produce $\hat{p}_i = \widehat{\Pr}(y_i = 1 \mid \mathbf{x}_i)$ for future applicants, such that (i) risky applicants are ranked above safe ones, (ii) $\hat{p}_i$ is numerically trustworthy, and (iii) an accept/reject rule built on $\hat{p}_i$ is profitable.

Two features of the problem shape the entire design. First, *leakage is the dominant failure mode*: a raw loan servicing extract contains dozens of post-origination fields (payments received, recoveries, hardship flags) that trivially predict the outcome, so the feature set must be restricted to an explicit origination-time whitelist. Second, *the deployment setting is out of time*: a model trained on past cohorts scores future applicants under a shifted economy, so a random train/test split flatters every model; splits must follow the calendar.

The scope is deliberately constrained: one consumer-credit market (Lending Club personal loans) as the main study with one independent robustness dataset, 36-month loans only so that every outcome is fully observed, origination-time features only, and a fixed loss-given-default assumption for the profit analysis.

2.2. Objectives

1. To implement an honest industry-standard baseline: a weight-of-evidence scorecard with information-value screening, logistic regression and points-to-double-odds scaling.
2. To implement three ML challengers (XGBoost, LightGBM, MLP) on the identical origination-time information set.
3. To design a leak-proof out-of-time evaluation protocol with discrimination (AUC, KS), calibration (Brier, ECE, reliability diagrams) and significance testing (DeLong, bootstrap).
4. To apply identical post-hoc recalibration (Platt, isotonic) to every model and measure whether the ML advantage survives it.
5. To evaluate all models by expected profit across accept/reject cutoffs under an explicit cash-flow model.
6. To verify robustness on an independent dataset and deliver a reproducible, modular codebase.

3. METHODOLOGY / APPROACH

3.1. Overall Workflow

The pipeline runs end to end from the raw loan export to a significance-tested comparison: (1) the raw export is reduced to an explicit whitelist of origination-time fields, and 36-month loans with resolved outcomes are retained; (2) loans are split by issue date into train (≤2013), validation (2014) and test (2015) vintages; (3) the scorecard and the three challengers are fit on the training vintages, with the validation vintage used for early stopping; (4) each model's raw PDs are recalibrated on the validation vintage by Platt scaling and isotonic regression, mimicking a bank recalibrating a deployed model on its most recent observed cohort; (5) raw and recalibrated PDs are scored on the 2015 test vintage for discrimination and calibration, with DeLong tests and bootstrap confidence intervals; (6) each model's PDs drive an accept/reject sweep whose realized profit is computed from the test loans' actual cash-flow outcomes. ML models are retrained across five seeds; every stage sees only information available at its point on the calendar.

3.2. Technical Details

Scorecard baseline. Each numeric feature is quantile-binned on the training data (missing values form their own bin); each categorical feature uses its levels with rare levels pooled. Bin j of a feature receives the weight of evidence

$$\mathrm{WoE}_j = \ln\left(\frac{g_j/g}{b_j/b}\right), \tag{1}$$

where g_j, b_j count goods and bads in the bin and g, b their totals (counts smoothed by 0.5). Features are screened by information value, $\mathrm{IV} = \sum_j (g_j/g - b_j/b)\,\mathrm{WoE}_j$, keeping $\mathrm{IV} \geq 0.02$ [3]. A logistic regression on the WoE-transformed features yields the PD, which is also expressed as an additive score scaled so that 600 points correspond to 19:1 good:bad odds and 20 points double the odds. This is the regulated-bank workhorse: every input enters through a monotone-ish transform and the final model is a readable sum of points.

ML challengers. XGBoost [6] and LightGBM [7] are gradient-boosted tree ensembles (2,000 trees maximum, learning rate 0.05, depth 5 / 31 leaves, row and column subsampling 0.8, early-stopped on validation AUC); the MLP is a two-layer network (64 and 32 units, ReLU, Adam, early stopping) on standardized inputs. All three consume the same design matrix, median-imputed numerics plus one-hot categoricals fit on the training vintages only, so performance differences come from the learner rather than the preprocessing. Each is retrained with five seeds and results are reported as mean ± standard deviation.

Calibration. A PD model is calibrated if, among loans assigned $\hat{p} \approx q$, a fraction q actually defaults. Post-hoc recalibration fits a monotone map on held-out data: Platt scaling [8] refits a one-parameter logistic on the raw model logit, and isotonic regression [9] fits a monotone step function. Both are fit on the 2014 validation vintage and applied to the 2015 test scores. Calibration is measured by the Brier score and the expected calibration error

$$\mathrm{ECE} = \sum_{k=1}^{10} \frac{n_k}{n} \left|\bar{y}_k - \bar{p}_k\right|, \tag{2}$$

over equal-frequency bins [10], alongside reliability diagrams.

Profit model. The lender accepts every applicant with $\hat{p}_i \leq c$ and the cutoff c is swept over $(0, 1)$. Realized profit per accepted loan uses the loan's actual outcome: a repaid 36-month loan earns its total interest, $36 \cdot \mathrm{installment}_i - \mathrm{principal}_i$ (undiscounted), and a charged-off loan loses $\mathrm{LGD} \times \mathrm{principal}_i$ with $\mathrm{LGD} = 0.65$, a standard unsecured-lending assumption. Profit is reported per 1,000 applicants. This is deliberately a simple, fully stated cash-flow model: it prices exactly the trade-off a PD cutoff controls, foregone interest against principal losses.

Significance. AUC differences on the shared test set are tested with DeLong's method for correlated ROC curves [11] (fast implementation of [12]), and confidence intervals for paired AUC and Brier differences use the percentile bootstrap over test loans (1,000 resamples).

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

Data. Lending Club accepted personal loans, public export 2007–2018Q4 [14]. Restricting to 36-month loans issued 2007–2015 guarantees every loan matured before the data ends, so no outcome is right-censored. After the origination-time whitelist and resolved-outcome filter: train 175,426 loans issued 2007–2013 (default rate 12.6%), validation 162,570 loans issued 2014 (13.7%), test 283,026

Table 1. Out-of-time leaderboard, Lending Club 2015 test vintage (283,026 loans). ML models: mean over five seeds (AUC std ≤ 0.006). Best value per column in bold.

Calib.	Model	AUC	KS	Brier	ECE
Raw	Scorecard	0.6767	0.2570	0.1213	0.0246
	XGBoost	**0.6850**	**0.2701**	0.1212	0.0338
	LightGBM	0.6843	0.2693	0.1213	0.0344
	MLP	0.6716	0.2542	0.1218	0.0275
Isotonic	Scorecard	0.6766	0.2564	0.1209	0.0175
	XGBoost	0.6848	0.2699	**0.1203**	0.0214
	LightGBM	0.6841	0.2688	0.1204	0.0218
	MLP	0.6714	0.2537	0.1212	**0.0172**

Table 2. Challengers versus the scorecard on the shared test set. ΔAUC on raw scores with DeLong p and bootstrap 95% CI; ΔBrier ($\times 10^3$, negative favours the challenger) after isotonic recalibration of both models.

Model	ΔAUC [95% CI]	DeLong p	$\Delta\text{Brier}_{\text{iso}}$
XGBoost	+0.0085 [0.0075, 0.0096]	$< 10^{-54}$	[−0.71, −0.53]
LightGBM	+0.0072 [0.0062, 0.0082]	$< 10^{-48}$	[−0.60, −0.43]
MLP	−0.0017 [−0.0031, −0.0002]	0.024	[+0.02, +0.24]

loans issued 2015 (14.9%). The steadily rising default rate across vintages is itself part of the experiment: it is exactly the drift a deployed model faces. Features: 17 numeric and 6 categorical origination-time fields (loan terms, applicant profile, credit-bureau snapshot); the scorecard's IV screen retains 11 of them, led by `sub_grade`, `int_rate`, `grade` and `fico`. The robustness dataset is the UCI Taiwan credit-card default set [13]: 30,000 clients, 23 features, stratified 60/20/20 split (it has no origination dates).

Environment. Windows 11, CPU only; Python 3.13 with `scikit-learn` 1.9, `xgboost` 3.3, `lightgbm` 4.6, `pandas` 3.0. All seeds fixed; the full study executes in about 18 minutes.

4.2. Results

Table 1 reports the test-vintage leaderboard: discrimination (AUC, KS) and calibration (Brier, ECE) for every model, raw and after isotonic recalibration; ML entries are mean ± std over five seeds. Platt scaling performs on par with isotonic on every model and is omitted for space (the full grid is in the repository). Figure 1 shows reliability diagrams before and after isotonic recalibration; every raw curve lies above the diagonal. Figure 2 shows the scorecard's score distributions for repaid versus charged-off loans, the classic KS picture. Figure 3 traces each model's realized profit per 1,000 applicants across PD cutoffs, and Table 3 lists the best operating points. Table 2 reports significance tests of each challenger against the scorecard. Table 4 gives the robustness leaderboard on the Taiwan dataset, where XGBoost leads the scorecard by 1.50 AUC points (DeLong $p = 2.8 \times 10^{-6}$). Figure 4 compares XGBoost's gain-based feature importance with the scorecard's information values; the rank correlation between the two is $\rho = 0.56$.

Table 3. Best operating points on the profit curves (isotonic-calibrated PDs). Profit in USD per 1,000 applicants; Δ is the gain over accepting everyone.

Model	Cutoff	Accept	Profit	Δ
Accept all	1.00	100.0%	705,171	–
Scorecard	0.24	93.3%	713,772	+8,602
MLP	0.25	94.0%	714,401	+9,230
LightGBM	0.25	93.7%	725,927	+20,756
XGBoost	0.23	90.7%	**728,517**	**+23,346**

Table 4. Robustness leaderboard, UCI Taiwan credit-card default (6,000 test clients, stratified split). AUC on raw scores; Brier and ECE after isotonic recalibration.

Model	AUC	$\text{Brier}_{\text{iso}}$	ECE_{iso}
Scorecard	0.7717	0.1363	**0.0127**
XGBoost	**0.7880**	**0.1330**	0.0162
LightGBM	0.7859	0.1334	0.0160
MLP	0.7662	0.1362	0.0171

5. DISCUSSION

The ML advantage is real, but it belongs to the trees. The boosted ensembles beat the scorecard decisively in statistical terms (Table 2): the DeLong p-values are astronomically small because 283,026 test loans give enormous power. Yet the economic size of the gap, +0.85 AUC points for XGBoost, is an order of magnitude smaller than the +8 to +15 point gaps sometimes quoted from random-split studies, and the MLP actually loses to the scorecard, echoing the conclusion of [5] that deep learning does not pay on tabular credit data. Two mechanisms compress the gap. First, the information set is already heavily distilled: `int_rate`, `grade` and `sub_grade` encode Lending Club's own underwriting model, so all four models largely re-rank applicants the platform has already risk-ranked (Fig. 4); the residual nonlinear signal available to the trees is limited. Second, the out-of-time protocol denies the flexible models the flattery of a random split, in which train and test share the same macroeconomic regime.

Calibration is a separate axis, and the scorecard wins it. Raw tree outputs are visibly miscalibrated (ECE 0.034 versus the scorecard's 0.025, Table 1), as expected for margin-maximizing ensembles [10]; the logistic scorecard is a probability model by construction. Isotonic recalibration narrows but does not erase this: the recalibrated scorecard still has the lower ECE (0.0175 versus 0.0214). Meanwhile the recalibrated trees keep their Brier advantage because Brier rewards discrimination as well as calibration. The reliability diagrams (Fig. 1) expose the deeper issue: all raw curves sit above the diagonal because the 2015 vintage defaulted more (14.9%) than the training vintages (12.6%). Recalibrating on 2014 corrects only the part of the drift already visible in 2014. No within-model fix can anticipate a level shift that has not happened yet; this is why regulators require ongoing monitoring rather than one-time calibration.

Profit concentrates the AUC gap at the cutoff. Accept-

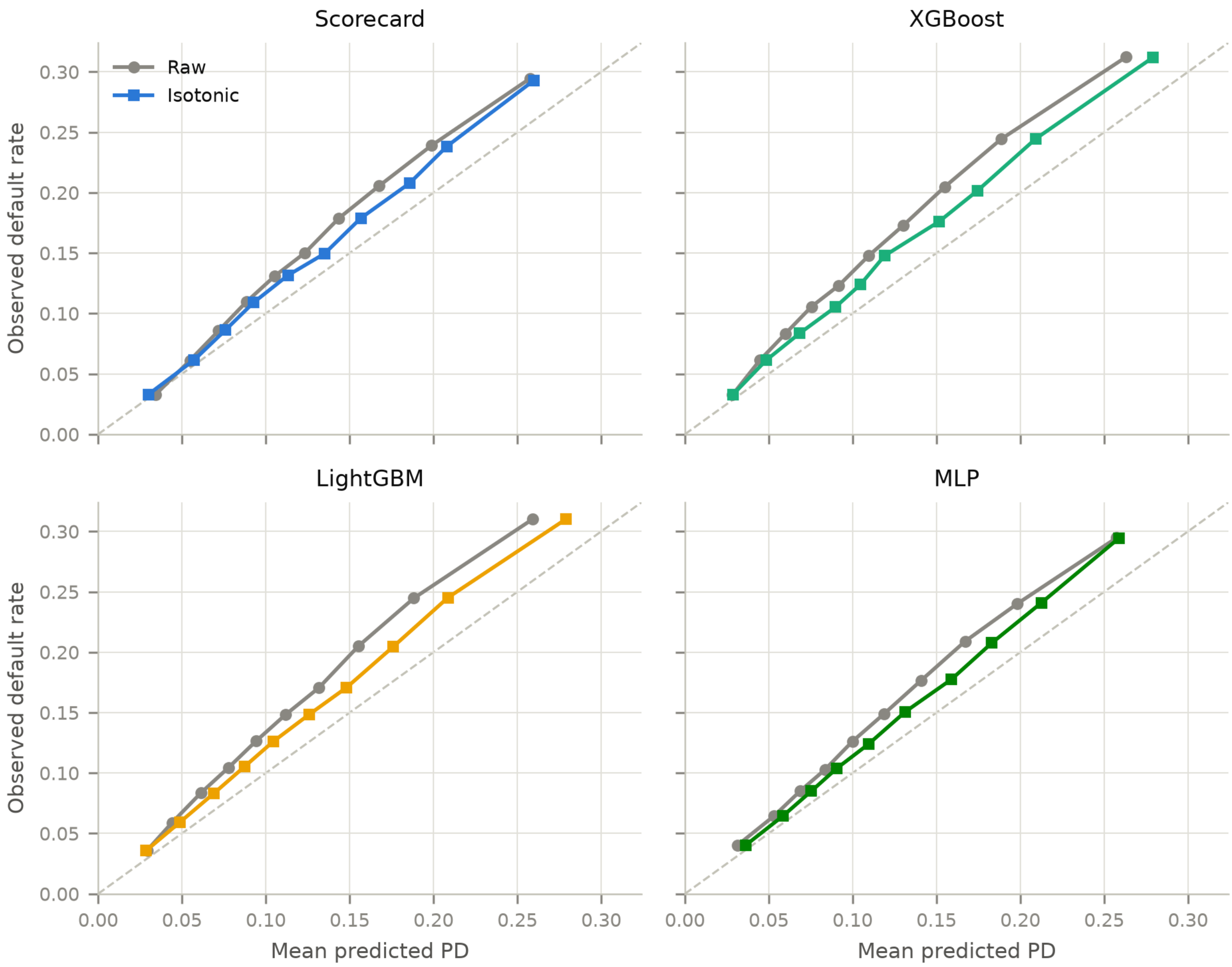

Fig. 1. Reliability diagrams on the 2015 test vintage (10 equal-frequency bins), raw (grey) versus isotonic-recalibrated (colour) predictions, per model. Every raw curve lies above the diagonal: the 2015 vintage defaulted more than the training vintages, so all models under-predict PD. Isotonic recalibration on the 2014 vintage removes most, but not all, of the bias.

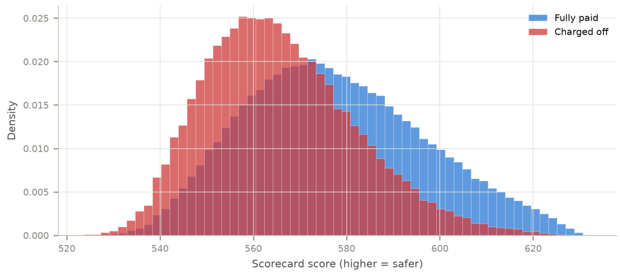

Fig. 2. Scorecard score distributions on the test vintage for repaid versus charged-off loans. The separation between the two distributions is the KS statistic (0.257).

ing everyone is already profitable (705 USD per applicant, Table 3) because interest income dominates in this portfolio; a PD model adds value only by pruning the riskiest tail. The scorecard's best policy rejects 6.7% of applicants for a +1.2% profit gain, while XGBoost rejects 9.3% for +3.3%; switching from scorecard to XGBoost is worth about 14,700 USD per 1,000 applicants (2.1%). The whole battle is fought in the zoomed panel of Fig. 3: everywhere else the models are interchangeable. This is the study's most practical observation: a 0.85-point AUC edge, statistically overwhelming, converts into a modest but genuine economic edge whose value scales linearly with volume, which is why large lenders adopt gradient boosting and small ones often rationally do not.

Trade-offs observed. The scorecard's WoE pipeline cost discrimination but bought near-calibration for free and full point-by-point readability, an attractive bargain under model-governance constraints. Isotonic versus Platt was immaterial here (both monotone maps preserve AUC and land within 10^{-3} Brier of each other), so the operational choice can be made on simplicity. Five-seed variance was negligible for the trees (±0.0005 AUC) but ten times larger for the MLP, another hidden cost of the neural option.

Limitations. (i) The data contain only *accepted* loans, so all models inherit Lending Club's selection; correcting for it (reject inference) is a known open problem [2]. (ii) The profit model is undiscounted, uses a fixed LGD = 0.65, and ignores partial recoveries on charged-off loans and prepayment; it prices the accept/reject margin, not portfolio economics. (iii) The platform's own risk grades were kept

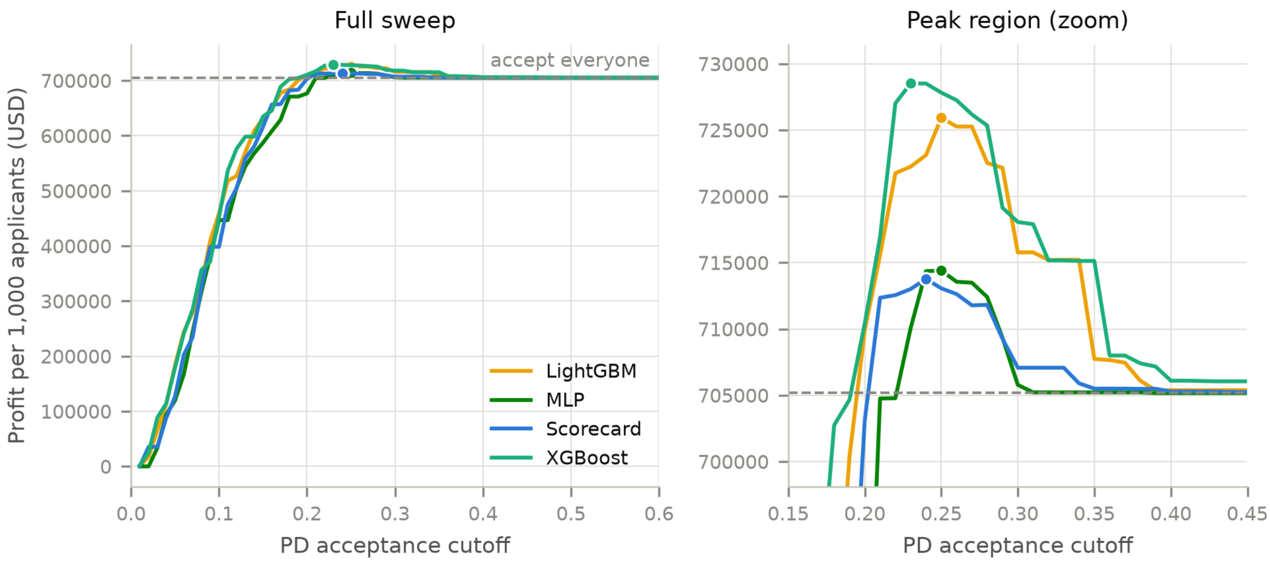

Fig. 3. Realized profit per 1,000 applicants versus PD acceptance cutoff, per model (isotonic-calibrated PDs), with best operating points marked and the accept-everyone benchmark dashed. Left: full sweep. Right: zoom on the peak region, where the boosted trees separate from the scorecard and MLP.

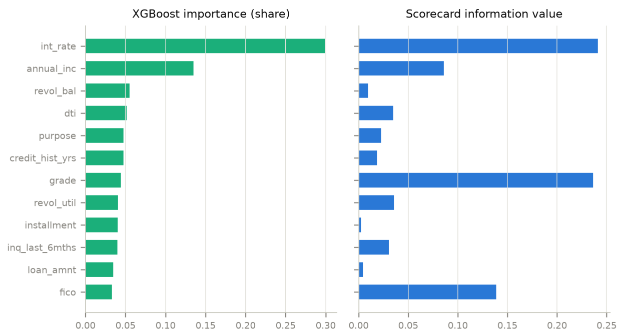

Fig. 4. What the two model families consider informative: XGBoost total-gain share (left) versus scorecard information value (right), top 12 features by XGBoost gain. Both agree that the platform's own risk pricing (`int_rate`, `grade`) dominates; they disagree on `sub_grade` versus `int_rate`, which carry nearly the same information in different encodings, and on `annual_inc`.

as features, which is realistic for an investor on the platform but understates what the challengers could add for a lender building from raw bureau data. (iv) One market and one test vintage; 2015 was benign, and a recession vintage could reorder the calibration findings. (v) The scorecard used equal-frequency WoE bins without monotonicity constraints or interaction terms; a production scorecard with expert binning would likely close part of the AUC gap.

6. CONCLUSION AND FUTURE WORK

Under a leak-proof out-of-time protocol on 621,022 Lending Club loans, gradient-boosted trees beat the industry-standard WoE scorecard on discrimination, on Brier score after identical recalibration, and on realized profit at the optimal accept/reject cutoff; the ranking reproduced on an independent dataset. The answer to the project's question is a qualified *yes*: the ML advantage is statistically unambiguous and survives the calibration and profit criteria the AUC-centric literature usually skips, but it is economically modest (+2.1% profit), it is delivered only by the boosted trees (the MLP significantly underperformed the scorecard), and the scorecard remained the best-calibrated model throughout.

The main learning outcomes were methodological. Leakage control dominated everything: reducing 151 raw columns to an origination-time whitelist was the single highest-leverage step in the project, and any mistake there would have produced impressive, meaningless AUCs. Second, evaluation criteria are not interchangeable: the model ranking changed depending on whether discrimination, calibration or profit was asked, so "which model is best" has no answer until the deployment question is fixed. Third, vintage drift is not a nuisance but the phenomenon itself: the 2014-fitted calibrators could not fully correct the 2015 default level, a small, honest demonstration of why PD models are monitored rather than trusted.

Future work: reject inference to de-bias the accepted-only sample; survival models for the *timing* of default rather than its occurrence; discounted cash-flow profit with empirical recoveries; monotonicity-constrained gradient boosting as a regulator-friendly middle ground; a fairness audit of the accept/reject policies; and stress evaluation on a recession vintage (2007–2008 originations).

7. ARTIFACTS AND DEMONSTRATIONS

- **Code repository:** `https://github.com/iitgF/T8`. Contains the modular `creditrisk` Python package (configuration, data and leakage filter, WoE engine, scorecard, ML models, calibration, metrics, profit analysis, pipeline) and the run script reproducing every number in this report.
- **LaTeX source (Overleaf):** `https://www.overleaf.com/project/6a524fb2acaf61e8bc3c5a8f`

- **Video demonstration:** *(placeholder, to be replaced with the final recording)*.

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